

1.

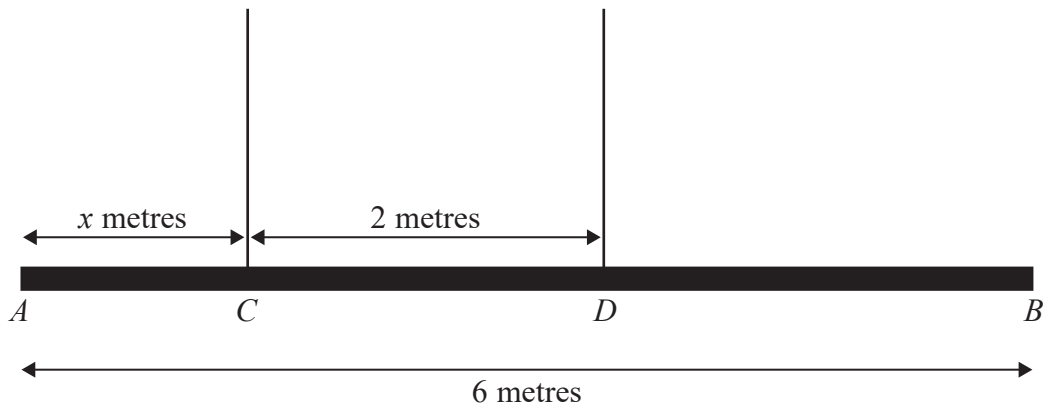


Figure 1

Figure 1 shows a beam AB with weight 24 N and length 6 m .

The beam is suspended by two light vertical ropes. The ropes are attached to the points C and D on the beam where $AC = x$ metres and $CD = 2\text{ m}$.

The tension in the rope attached to the beam at C is double the tension in the rope attached to the beam at D .

The beam is modelled as a uniform rod, resting horizontally in equilibrium.

Find

(i) the tension in the rope attached to the beam at D .

(ii) the value of x .

2.

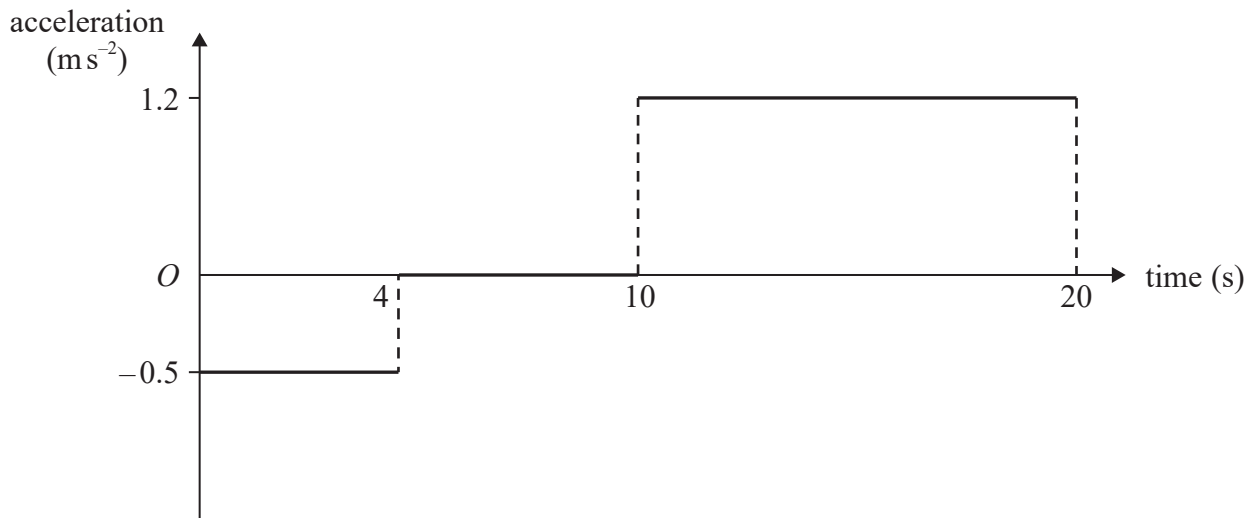


Figure 2

Two fixed points, A and B , are on a straight horizontal road.

The **acceleration-time** graph in Figure 2 represents the motion of a car travelling along the road as it moves from A to B .

At time $t = 0$, the car passes through A with speed 8 m s^{-1}

At time $t = 20 \text{ s}$, the car passes through B with speed $v \text{ m s}^{-1}$

(a) Show that $v = 18$

(3)

(b) Sketch a speed-time graph for the motion of the car from A to B .

(3)

(c) Find the distance AB .

(4)

3. A hammer is used to hit a tent peg into soft ground.

The hammer has mass 1.8 kg and the tent peg has mass 0.2 kg .

The hammer and tent peg are both modelled as particles and the impact is modelled as a direct collision.

Immediately before the impact, the tent peg is stationary and the hammer is moving vertically downwards with speed 10 m s^{-1}

Immediately after the impact, the hammer and tent peg move together, vertically downwards, with the **same** speed $v\text{ m s}^{-1}$

- (a) Find the value of v (2)

- (b) Find the magnitude of the impulse exerted on the tent peg by the hammer, stating the units of your answer. (3)

The ground exerts a constant vertical resistive force of magnitude R newtons, bringing the hammer and tent peg to rest after they travel a distance of 12 cm .

- (c) Find the value of R . (5)
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4. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively.]

A particle P moves with constant acceleration $(-\lambda\mathbf{i} + 2\lambda\mathbf{j})\text{ m s}^{-2}$, where λ is a positive constant.

At time $t = 0$, the velocity of P is $(5\mathbf{i} - 8\mathbf{j})\text{ m s}^{-1}$

- (a) Find the velocity of P when $t = 5\text{ s}$, giving your answer in terms of $\mathbf{i}$, $\mathbf{j}$ and λ .

(2)

The speed of P when $t = 5\text{ s}$ is 13 m s^{-1}

- (b) Show that

$$25\lambda^2 - 42\lambda - 16 = 0$$

(3)

- (c) Find the direction of motion of P when $t = 4\text{ s}$, giving your answer as a bearing to the nearest degree.

(5)

5.

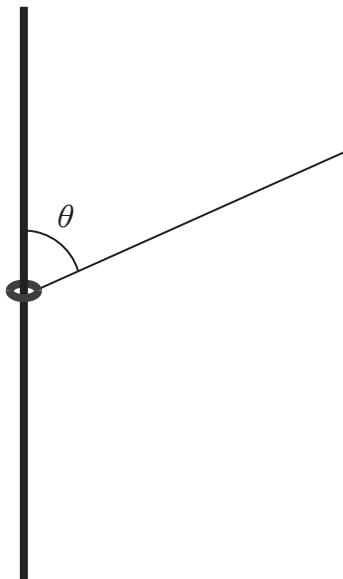


Figure 3

A small ring of mass 0.2 kg is attached to one end of a light inextensible string.

The ring is **threaded** onto a fixed rough vertical rod.

The string is taut and makes an angle θ with the rod, as shown in Figure 3,
where $\tan \theta = \frac{12}{5}$

Given that the ring is in equilibrium and that the tension in the string is 10 N ,

(a) find the magnitude of the frictional force acting on the ring,

(3)

(b) state the direction of the frictional force acting on the ring.

(1)

The coefficient of friction between the ring and the rod is $\frac{1}{4}$

Given that the ring is in equilibrium, and that the tension in the string, T newtons, can now vary,

(c) (i) find the minimum value of T

(ii) find the maximum value of T

(8)

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At 12:00, a ship P sets sail from a harbour with position vector $(15\mathbf{i} + 36\mathbf{j})$ km.

At 12:30, P is at the point with position vector $(20\mathbf{i} + 34\mathbf{j})$ km.

Given that P moves with constant velocity,

- (a) show that the velocity of P is $(10\mathbf{i} - 4\mathbf{j})\text{ km h}^{-1}$

(2)

At time t hours after 12:00, the position vector of P is $\mathbf{p}$ km.

- (b) Find an expression for $\mathbf{p}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and t .

(2)

A second ship Q is also travelling at a constant velocity.

At time t hours after 12:00, the position vector of Q is given by $\mathbf{q}$ km, where

$$\mathbf{q} = (42 - 8t)\mathbf{i} + (9 + 14t)\mathbf{j}$$

Ships P and Q are modelled as particles.

If both ships maintained their course,

- (c) (i) verify that they would collide at 13:30

(ii) find the position vector of the point at which the collision would occur.

(4)

At 12:30 Q changes speed and direction to avoid the collision.

Ship Q now travels due north with a constant speed of 15 km h^{-1}

Ship P maintains the same constant velocity throughout.

- (d) Find the exact distance between P and Q at 14:30

(7)

7.

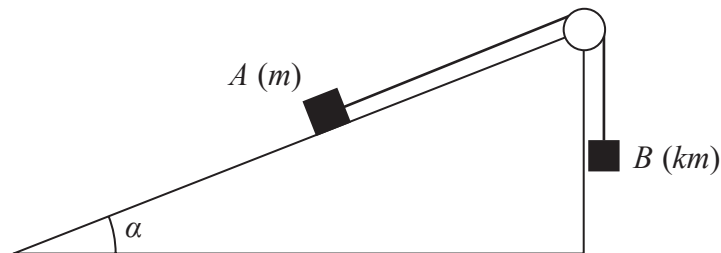
**Figure 4**

Figure 4 shows a block A of mass m held at rest on a rough plane. The plane is inclined at an angle α to the horizontal and the coefficient of friction between the block and the plane is μ .

One end of a light inextensible string is now attached to A . The string passes over a small smooth pulley which is fixed at the top of the plane.

The other end of the string is attached to a block B of mass km .

Block B hangs vertically below the pulley, with the string taut.

The string from A to the pulley lies along a line of greatest slope of the plane.

Both A and B are modelled as particles.

When the system is released from rest, A moves up the plane and the tension in the string is $\frac{4mg}{3}$

Given that $\mu = \frac{1}{3}$ and $\tan \alpha = \frac{7}{24}$

- (a) (i) find the magnitude of the acceleration of A , giving your answer in terms of g ,
- (ii) find the value of k .

(9)

- (b) Find the magnitude of the resultant force exerted on the pulley by the string, giving your answer in terms of m and g .

(4)